MATHS MANTHAN
• TUITION CLASSES •
e-mail: contact@mathsmanthan.co.in | Website:- mathsmanthan.co.in

Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – WEEKLY TEST 03
Chapter 13: Statistics

Time: 45 Minutes
MM: 20

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in **Test Paper:** Weekly Test 03

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. In the formula $\bar{x} = a + \dfrac{\sum f_i d_i}{\sum f_i}$ for finding the mean of grouped data, d_i are deviations from a of: [1]

(A) lower limits of classes (B) upper limits of classes
(B) mid-points of classes (D) frequencies of classes

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) mid-points of classes

Justification: In the assumed mean method, deviations d_i are calculated as $d_i = x_i - a$, where x_i represents the mid-point (class mark) of the i-th class interval and a is the assumed mean.

Q2. If the difference between mode and median of a data is 24, then the difference between median and mean is: [1]

(A) 8 (B) 12 (C) 24 (D) 36

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 12

Justification: By empirical formula:

$$\text{Mode} = 3\text{Median} - 2\text{Mean} \implies \text{Mode} - \text{Median} = 2(\text{Median} - \text{Mean})$$

Given Mode − Median = 24:

$$24 = 2(\text{Median} - \text{Mean}) \implies \text{Median} - \text{Mean} = \frac{24}{2} = \mathbf{12}.$$

$$\boxed{\text{Median} - \text{Mean} = \mathbf{12}}$$

Q3. Consider the following frequency distribution: [1]

Class	$0-5$	$6-11$	$12-17$	$18-23$	$24-29$
Frequency	13	10	15	8	11

The upper limit of the median class is:

(A) 17 (B) 17.5 (C) 18 (D) 18.5

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 17.5

Justification: Converting the inclusive discontinuous classes into continuous classes by subtracting and adding 0.5 ($\frac{6-5}{2} = 0.5$):

- $-0.5 - 5.5$: $f = 13$, $cf = 13$
- $5.5 - 11.5$: $f = 10$, $cf = 23$
- $11.5 - 17.5$: $f = 15$, $cf = 38$
- $17.5 - 23.5$: $f = 8$, $cf = 46$
- $23.5 - 29.5$: $f = 11$, $cf = 57$

Total frequency $N = 57 \implies \frac{N}{2} = \frac{57}{2} = 28.5$. The cumulative frequency just greater than 28.5 is 38, which corresponds to the class interval $\mathbf{11.5 - 17.5}$.
Hence, the upper limit of the continuous median class is **17.5**.

Q4. If the mean of the numbers $27+x, 31+x, 89+x, 107+x, 156+x$ is 82, then the mean of $130+x, 126+x, 68+x, 50+x, 1+x$ is: [1]

(A) 75 (B) 80 (C) 82 (D) 70

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) 75

Justification: Mean of the first 5 numbers:

$$\bar{x}_1 = \frac{(27+31+89+107+156)+5x}{5} = \frac{410+5x}{5} = 82+x$$

Given $\bar{x}_1 = 82 \implies 82+x = 82 \implies x = 0$.
Mean of the second 5 numbers with $x = 0$:

$$\bar{x}_2 = \frac{130+126+68+50+1+5(0)}{5} = \frac{375}{5} = \mathbf{75}.$$

Required Mean = **75**

Q5. **Assertion (A):** If each observation of a data set is increased by 5, the arithmetic mean is also increased by 5.
Reason (R): The sum of deviations of a set of observations from their mean is always non-zero. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (C) (A) is true but (R) is false.

Justification: If each observation x_i is increased by a constant $c = 5$, the new mean is $\frac{\sum(x_i+5)}{n} = \frac{\sum x_i}{n} + 5 = \bar{x} + 5$. Hence, Assertion (A) is true.
However, by the fundamental property of the arithmetic mean, $\sum(x_i - \bar{x}) = 0$ (the sum of deviations from the mean is always zero, never non-zero). Hence, Reason (R) is false.

SECTION B: Very Short Answer Type — MATHS MANTHAN | (1 × 2 = 2 Marks)

Q6. For a frequency distribution, given $\sum f_i d_i = 36$, $\sum f_i = 40$, and assumed mean $a = 25$. Calculate the arithmetic mean. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Formula Declaration [1 Mark]
By the Assumed Mean Method:

$$\bar{x} = a + \frac{\sum f_i d_i}{\sum f_i}$$

Step 2: Substitution & Calculation [1 Mark]
Given $a = 25$, $\sum f_i d_i = 36$, and $\sum f_i = 40$:

$$\bar{x} = 25 + \frac{36}{40} = 25 + \frac{9}{10} = 25 + 0.9 = 25.9$$

$$\boxed{\text{Arithmetic Mean } (\bar{x}) = \mathbf{25.9}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (3 × 3 = 9 Marks)

Q7. Find the mode of the following distribution of daily wages: [3]

Wages (Rs)	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
Workers	4	5	12	2	2

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying Modal Class & Parameters [1 Mark]

The maximum frequency is 12, corresponding to the class **200 – 250**.

$\therefore$ Modal Class = 200 – 250.

Lower limit (l) = 200, $f_1 = 12$, $f_0 = 5$ (preceding frequency), $f_2 = 2$ (succeeding frequency), class width (h) = 50.

Step 2: Mode Formula & Substitution [1 Mark]

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

$$\text{Mode} = 200 + \left(\frac{12-5}{2(12)-5-2}\right) \times 50 = 200 + \left(\frac{7}{24-7}\right) \times 50 = 200 + \frac{350}{17}$$

Step 3: Calculation & Final Boxed Answer [1 Mark]

$$\text{Mode} = 200 + 20.588... \approx 220.59 \text{ Rs} \quad \left(\text{or } 220\frac{10}{17} \text{ Rs}\right)$$

$$\boxed{\text{Mode of Daily Wages} = \text{Rs } \mathbf{220.59} \quad \left(\text{or Rs } \mathbf{220\frac{10}{17}}\right)}$$

Q8. Calculate the mean of the following data using the Step-Deviation Method: [3]

Class Interval	25 – 35	35 – 45	45 – 55	55 – 65	65 – 75
Frequency	6	10	8	12	4

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Step-Deviation Table [1.5 Marks]

Let Assumed Mean $a = 50$, class width $h = 10$, $u_i = \dfrac{x_i - 50}{10}$.

Class Interval	**Frequency (f_i)**	**Class Mark (x_i)**	$u_i = \frac{x_i - 50}{10}$	$f_i u_i$
25 – 35	6	30	−2	−12
35 – 45	10	40	−1	−10
45 – 55	8	50	0	0
55 – 65	12	60	+1	+12
65 – 75	4	70	+2	+8
Total	$\sum f_i = 40$			$\sum f_i u_i = -2$

Step 2: Formula Declaration & Substitution [1 Mark]

$$\bar{x} = a + \left(\frac{\sum f_i u_i}{\sum f_i}\right) \times h = 50 + \left(\frac{-2}{40}\right) \times 10 = 50 - \frac{2}{4} = 50 - 0.5 = 49.5$$

Step 3: Final Boxed Result [0.5 Mark]

$$\boxed{\text{Mean of the data } (\bar{x}) = \mathbf{49.5}}$$

Q9. Convert the discontinuous classes into continuous classes and calculate the mode: [3]

Class	11 – 20	21 – 30	31 – 40	41 – 50	51 – 60
Frequency	8	15	20	12	5

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Continuous Class Conversion & Modal Class [1 Mark]

$$\text{Adjustment factor} = \frac{21-20}{2} = 0.5.$$

Given Class	Continuous Class	Frequency (f_i)
$11-20$	$10.5-20.5$	8
$21-30$	$20.5-30.5$	15
$31-40$	$\mathbf{30.5-40.5}$	**20** (Modal Class)
$41-50$	$40.5-50.5$	12
$51-60$	$50.5-60.5$	5

Step 2: Formula Declaration & Substitution [1 Mark]

Here, modal class $= 30.5-40.5$, $l = 30.5$, $f_1 = 20$, $f_0 = 15$, $f_2 = 12$, $h = 10$.

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h = 30.5 + \left(\frac{20-15}{2(20)-15-12}\right) \times 10 = 30.5 + \frac{50}{13}$$

Step 3: Calculation & Final Result [1 Mark]

$$\text{Mode} = 30.5 + 3.846 \approx 34.35 \quad \left(\text{or } 34\frac{9}{26}\right)$$

$$\boxed{\text{Mode} = \mathbf{34.35} \quad \left(\text{or } \mathbf{34\frac{9}{26}}\right)}$$

SECTION D: Case Study Based Question

MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. CASE STUDY: METEOROLOGICAL DEPARTMENT SEASONAL MONSOON RAINFALL RECORD [4 Marks]

The India Meteorological Department (IMD) observed seasonal rainfall records across various agro-climatic sub-divisions during the monsoon season to assist in agricultural crop planning and water resource management.

Rainfall (mm)	$200-400$	$400-600$	$600-800$	$800-1000$	$1000-1200$	$1200-1400$	$1400-1600$
Sub-divisions	2	4	6	10	8	4	2

(i) State the modal rainfall class and find its class mark. [1]

(ii) Calculate the mode of the seasonal rainfall across the sub-divisions. [2]

(iii) Find how many sub-divisions received less than 1000 mm of rainfall. [1]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Subpart (i): Modal Rainfall Class & Class Mark [1 Mark]

The maximum frequency is 10, which corresponds to the class **800 − 1000**.
∴ Modal Class = 800 − 1000 mm.
Class mark of the modal class:

$$\text{Class Mark} = \frac{800 + 1000}{2} = \frac{1800}{2} = \mathbf{900}\ \text{mm}.$$

$$\boxed{\text{Modal Class} = \mathbf{800 - 1000}\ \text{mm}, \quad \text{Class Mark} = \mathbf{900}\ \text{mm}}$$

Subpart (ii): Mode Calculation [2 Marks]

For modal class 800 − 1000: $l = 800$, $f_1 = 10$, $f_0 = 6$, $f_2 = 8$, $h = 200$.
Formula:

$$\text{Mode} = l + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h \text{ [1 Mark]}$$

$$\text{Mode} = 800 + \left(\frac{10 - 6}{2(10) - 6 - 8}\right) \times 200 = 800 + \left(\frac{4}{20 - 14}\right) \times 200 = 800 + \left(\frac{4}{6}\right) \times 200$$

$$\text{Mode} = 800 + \frac{400}{3} = 800 + 133.33 = \mathbf{933.33}\ \text{mm} \quad \left(\text{or } 933\frac{1}{3}\ \text{mm}\right) \text{ [1 Mark]}$$

$$\boxed{\text{Mode of Seasonal Rainfall} = \mathbf{933.33}\ \text{mm} \quad \left(\text{or } \mathbf{933\frac{1}{3}}\ \text{mm}\right)}$$

Subpart (iii): Sub-divisions Receiving Less Than 1000 mm [1 Mark]

The sub-divisions receiving less than 1000 mm comprise classes [200 − 400], [400 − 600], [600 − 800], and [800 − 1000]:

$$\text{Total Sub-divisions} = 2 + 4 + 6 + 10 = \mathbf{22}.$$

$$\boxed{\text{Number of sub-divisions with rainfall } < 1000\ \text{mm} = \mathbf{22}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★